report spektroscopy 1

1) Introduction

2) Experiment

3) Results

$k0$ denotes the k-value which corresponds to the 0-th order. The measured value is $k0 = 950$. This value is wrong. More details are in the discussion section.

the first measurement is to fit the wavelength of the middle pixel position in dependance of dial position (k-value). The results are the following:

wavelength [nm]	388.8	447.1	471.3	492.1	504.7	667.8	706.5
k-value	1310	1131	1101	1072	1070	849	800

The first column contains the wavelength in nm of the measured light-strips. The value is the literature value. The second column contains the measurements. The measurement is the k-value. This is light from the He-Lamp.

the second measurement is for a fixed k-value and measures data to fit the wavelength in dependance of pixel position the results are the following:

wavelength [nm]	492.1	501.5	504.7
pixel position	475	802	920

The first column is again the known wavelengths of the measured light-strips in nm. the second column is the pixel position on the image. the image is 1440px wide, therefore the pixel position p has $0 \leq p < 1440$

the third data-set is from the Hydrogen-Lamp to measure the wavelengths of the Balmer-Series.

n	3	4	5	6
k-value	860	1080	1112	1120

the first column is the Balmer-Series. The are the corresponding measurements of the k-values.

4) Data Analysis

The fit function used is of the form $\lambda(p,k) = bk + a + \frac{p-p_{\text{cent}}}{d}$ we used this function beacause our k0-value was wrong. Therefore we now use three fit-parameter instead of two like in the manualll. with the first data-set we obtained values for a and b. we used scipy.curve_fit for the regression. For the first data-set we used the fit $\lambda_{\text{c}}\ (k) = bk + a$ since $p = p_{\text{cent}}$. This gives the values $a = 1220, b = -0.663$ **insert plot**

for the second data-set the fit function is $\lambda(p, k_{\text{fix}}) = bk_{\text{fix}} + a + \frac{p-p_{\text{cent}}}{d}$. this is the final fit function but with a fixed $k = k_{\text{fix}}$ and a, b from above. this gives $d = 42.6$. **insert plot**

from the fit we calculate the lattice constant g as $g = \frac{1}{d} = 5.87e-05$.

with the complete fit function we obtained the following wavelengths of the Balmer-Series through the fit function: $\lambda 3 = 650\ \text{nm}, \lambda 4 = 504\ \text{nm}, \lambda 5 = 483\ \text{nm}, \lambda 6 = 477\ \text{nm}$.

from the average of the obtained wavelengths we calculate the Rydberg-Frequency as $R_{\text{H}} = 3.07e15$ therefore the limit of the Balmer-Series is $\lambda_{\text{min}} = 391\ \text{nm}$. From the Rydberg-Frequency we continue to calculate the ionisation energy as $E_{\text{ion}} = \frac{hR_{\text{H}}}{e} = 12.7\ \text{eV}$ where h is Plancks constant and e is the proton charge.

5) Discussion

The $k0$-value measured in the experiment is $k0 = 950$. This measurment does not correspond to the actual $k0$-vlaue but to the 587.5 nm wavelength light. This wavelength is yellow and was therefore misinterpreted as the 0-th order. In order to correct for this mistake, we used for the first fit-function $\lambda_c\ (k) = bk + a$ instead of $\lambda_c\ (k) = b(k - k0)$. The latter function is used in the manual.

The measurements of the Balmer-Series have given:

literature value [nm]	656	486	434	410
experimental value [nm]	650	504	483	477

Also the experimental values from the Rydberg-Frequency, ionisation energy and the limit of the Balmer-Series deviate from the literature values.

	Rydberg-Frequency [Hz]	ionisation energy [eV]	limit of Series [nm]
literature value	3.29 e 15	13.6	365
experimental value	3.07 e 15	12.7	391

The errors from the Rydberg-Frequency, ionisation energy and limit of the Balmer-Series directly propagate from the errors in measurement of the H-Lines. The values obtained from the H-Lines come from the fit-function and the measured k-values for each H-Line. Both of them must have their own errors, which contribute. We assume that the main errors comes from the fit-function.

6) Conclusion